

JadeSpectrogram2 Manual

the musical spectrogram

J. Bitzer, License: CC-BY 4.0

October 6, 2026, for JadeSpectrogram2 2.4.6 (manual build 30)

1 Introduction

Dear User,

a spectrogram is not the sexiest of all plug-ins, but sometimes very versatile. This incarnation comes as a plug-in and as a standalone application and is highly flexible in terms of its analysis parameters. You can tune it for your purposes.

JadeSpectrogram2 is the new version of the JadeSpectrogram, with a new code base, and with version 2.0 it became *the musical spectrogram*: a keyboard overlay shows which note a frequency belongs to, and a grid from the tempo of your DAW shows bars and beats. It only analyses: the audio passes through unchanged and without any latency, so you can put it anywhere in your signal chain.

I hope you will enjoy using it and gaining knowledge for acoustical or musical problems.

Have fun!

Joerg aka audio-dsp.

2 Installation

The downloaded file is a standard zip file. It contains a plug-in directory (.vst3), an AU file (.component, macOS only), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place for VST3 / AU plug-ins and let your DAW rescan its plug-ins.

2.1 Windows

For Windows™ the standard VST3-directory is:

C:\Program Files\Common Files\VST3

2.2 Mac VST

If you use the VST3 plugin, copy the .vst3 directory to

/Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3

2.3 Mac AudioUnit

If you use the AudioUnit plugin, copy the .component file/directory to

`/Users/yourUSERNAME/Library/Audio/Plug-Ins/Components`

The macOS version runs natively on Apple Silicon and Intel Macs. It is not signed with an Apple Developer ID yet. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g.

```
xattr -cr ~/Library/Audio/Plug-Ins/VST3/JadeSpectrogram2.vst3
```

2.4 Linux

For Linux copy the full directory to:

`/home/yourUSERNAME/.vst3/`

2.5 Standalone

The Standalone version runs without a DAW: it analyses the audio input of your sound card. Choose your audio device and input channels in its settings.

2.6 (De)-Installation

At the first start, the plug-in creates a (still empty) folder for presets. If you remove the plug-in, you can delete this folder as well:

Windows:

`C:\Users\yourUSERNAME\AppData\Roaming\Jade_Hochschule\JadeSpectrogram2`

Mac:

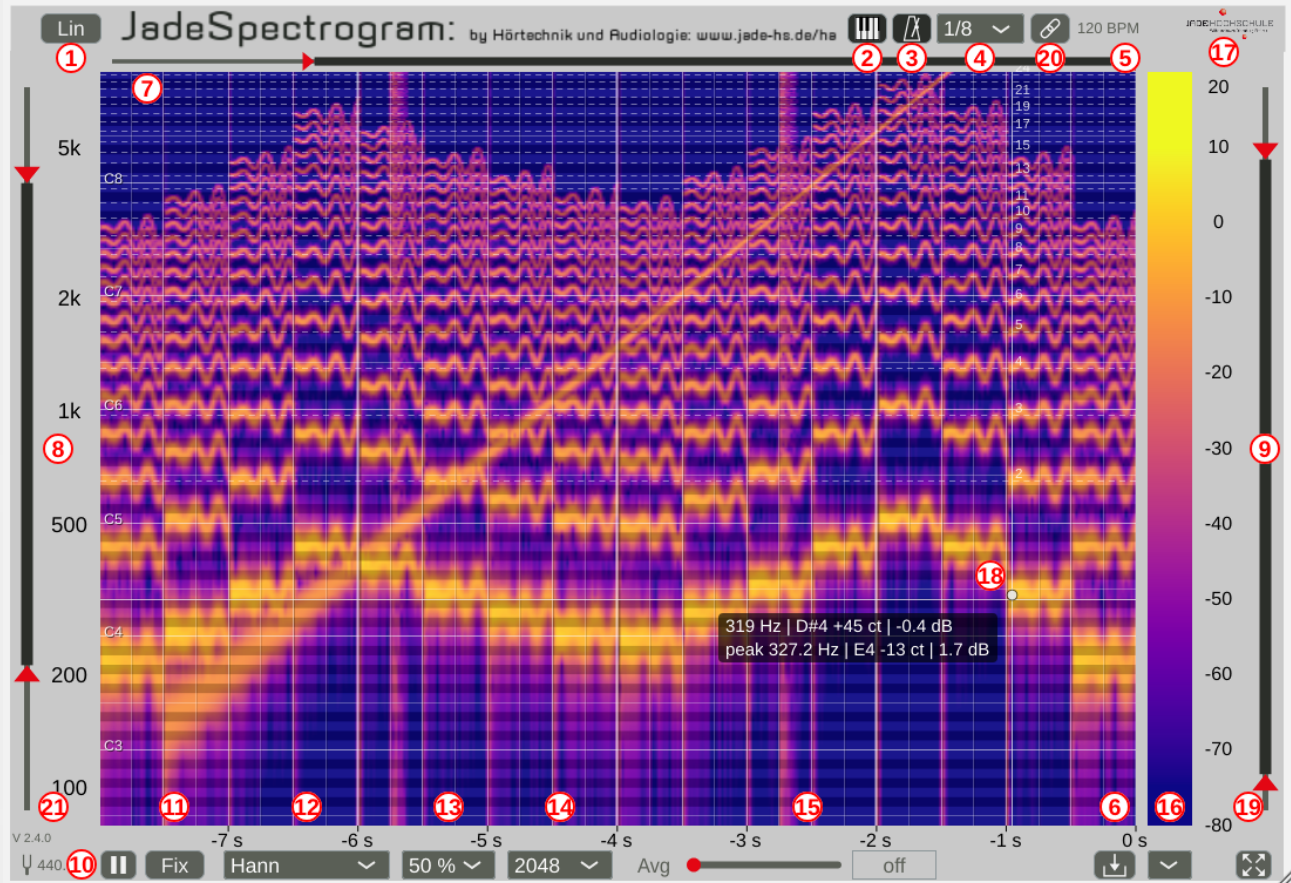
`/Users/yourUSERNAME/Library/Audio/Presets/Jade_Hochschule/JadeSpectrogram2`

Linux:

`/home/yourUSERNAME/.config/Jade_Hochschule/JadeSpectrogram2`

3 How to use

JadeSpectrogram2 has the analysis display in the middle, the axes and range sliders around it, the settings in the bottom row and the overlays and the export in the title bar.



The example shows a short melody at 120 BPM with its harmonics (with vibrato), an exponential sine sweep and two noise bursts, on the logarithmic axis with the keyboard overlay and the BPM grid (resolution 1/8, every eighth note). The crosshair (18) points to the fundamental of a note (E4), with the peak marked by a dot and the harmonic cursor (Shift) on.

3.1 Title bar (1–5, 17, 20)

- **Lin / Log (1):** switches the frequency axis between linear and logarithmic (the logarithmic axis starts at 20 Hz at the lowest). The button shows what a click will do. The frequency slider (8) follows the scale of the axis.
- **Keyboard overlay (2):** a transparent piano roll over the spectrogram. Every semitone gets a band from a quarter tone below to a quarter tone above the note (A4 = reference pitch (21), default 440 Hz); black keys are darker, lines mark where two white keys meet (E/F, B/C), stronger at every C. The note names appear at every C, and at every note when you zoom in far enough. The band boundaries are the same as in the readout of the crosshair (18): if the crosshair shows F#4, the mouse is in the F#4 band. On the linear axis, bands that would be thinner than a few pixels (low frequencies) fade out.

- **BPM grid (3)**: vertical lines at the bars (strong), the beats and finer note values, from the tempo and the position of your DAW. The lines stay on the beats also after tempo changes, jumps and loops (also at the loop start when the host jumps back in the middle of an audio block); while the transport is stopped the musical position does not advance, so there are no lines. Finer lines are left out where they would be closer than 5 pixels.
- **Grid resolution (4)**: note values as in the grid of a DAW, counted from the start of each bar: 1 shows only the bar lines, 1/2, 1/4, 1/8 and 1/16 a line every half, quarter, eighth or sixteenth note (default 1/4: in 4/4 a line at every beat). Bar lines are the strongest, then the beats of the time signature (a quarter note in 4/4, an eighth in 6/8). If the lines would be closer than 5 pixels, only the beats or only the bars are drawn. Only visible while the grid is on.
- **Sync / Free (20)**: where the tempo of the grid comes from. **Synced** (closed chain, default): the tempo and position of the host, the lines show where the beats really were. While the display is paused and you change the tempo in the host, the grid is redrawn at the new tempo from the last bar line, e.g. to find the tempo of the paused audio with the tempo control of your DAW. **Free** (broken chain): your own tempo, for tempo analysis, also with a stopped transport, while paused, and in the Standalone application. When you switch to Free, the grid takes over the current host tempo and continues the host grid without a jump. **Alt+click** in the display puts a bar line at that time; the grid counts from there, so you can put bar 1 on a downbeat and then adjust the tempo until the lines sit on the onsets. The free grid uses the last time signature of the host (otherwise 4/4).
- **BPM value (5)**: synced: the tempo of the host (it follows the host also while the transport is stopped or the display is paused, as long as the host processes audio); *no tempo* if there is none (e.g. in the Standalone application). Free: your tempo, 20 to 300 BPM in steps of 0.1. Drag up or down to change it (0.25 BPM per pixel; with Shift 0.02 BPM per pixel), use the mouse wheel (± 1 BPM, with Shift ± 0.1), or double-click and type a value. The two small triangles show that the value can be changed.
- **About box (17)**: click on the Jade Hochschule logo, click anywhere to close it.

3.2 Range sliders (7, 8, 9)

Each range slider has two thumbs (red triangles). Drag a thumb to move one end of the range, or drag between the thumbs to move the whole range; its width stays the same. The overview button (19) sets all three to their full range.

- **Time zoom (7)**: the visible part of the 10 seconds in memory (at least 0.2 s).
- **Frequency range (8)**: the displayed frequencies, 1 Hz to 20 kHz on the linear axis, 20 Hz to 20 kHz on the logarithmic axis (at most half the sampling rate). This does not change the analysis.
- **Colour range (9)**: the levels that the colourmap spans (-80 to $+20$ dB), for more contrast in the level region you are most interested in.

3.3 Bottom row (6, 10–16, 19, 21)

- **Reference pitch (21)**: the frequency of A4 (tuning fork), default 440 Hz, from 380 to 480 Hz in steps of 0.1 Hz. It sets the note names of the keyboard overlay (2) and of the crosshair (18), e.g. 415 Hz for baroque pitch or 442/443 Hz for most European orchestras; the analysis and the frequency axis do not change. Drag up or down (0.1 Hz per pixel, with Shift 0.01 Hz), use the mouse wheel (± 1 Hz, with Shift ± 0.1), or double-click and type a value.
- **Pause (10)**: stops the analysis, the display freezes for a closer look (play symbol, highlighted). The audio still passes through. Click again to continue.
- **Display mode (11)**: by default the spectrogram scrolls from right to left, the newest data is at the right edge. **Fix** freezes the image in place; a red line then runs from left to right and overwrites the oldest data. **Run** scrolls again. The button shows what a click will do.
- **Window (12)**: the short-time Fourier transform weights each block with a window: Rectangular, Hann, Hamming, Blackman-Harris, Flat Top and Hann-Poisson. When in doubt use Hann; Flat Top is best for reading exact levels of sinusoids, Blackman-Harris shows weak components next to strong ones.
- **Overlap (13)**: 50 % or 75 % overlap of the blocks. 75 % gives twice as many columns and a smoother time axis, useful with large FFT sizes (table 1).
- **FFT size (14)**: 512 to 8192. Use higher values for detailed frequency analysis and lower values for timing analysis (table 1). Changing the size or the overlap restarts the display.
- **Averaging (15)**: exponential averaging along time with the shown time constant; steady sounds become smoother, short events fade. The lowest part of the slider is *off*; above it, the time constant starts at two columns (the scale follows sampling rate, FFT size and overlap) and goes up to 2 s.
- **Colourmap (16)**: Mono, BW, Hot, Rainbow, Viridis, Plasma, Jade (default: Plasma). Viridis and plasma are equally spaced in their brightness perception and suitable for colour-blind people.
- **Export (6)**: saves the visible spectrogram with the frequency and time axis and the colour bar as a PNG file (twice the screen resolution, without the crosshair).
- **Overview (19)**: sets the frequency (8), time (7) and colour range (9) to their full range, to get the overview back after zooming in.

3.4 Analysis display, time axis and crosshair (18)

The main area shows the last 10 seconds of the analysed signal (or the part you selected with the time zoom) on the x-axis vs frequency on the y-axis; the colours show the power at each frequency. For a stereo signal, the power spectra of both channels are averaged.

The time axis below the display shows seconds: in the scrolling mode relative to now (-10 s ... 0 s at the right edge), in the Fix mode as sweep time (0 s at the left edge ... 10 s at the right), so the labels stay still while the red line shows the current position.

FFT size	block length	frequency spacing	new column (50 %)	new column (75 %)
512	10.7 ms	93.8 Hz	5.3 ms	2.7 ms
1024	21.3 ms	46.9 Hz	10.7 ms	5.3 ms
2048	42.7 ms	23.4 Hz	21.3 ms	10.7 ms
4096	85.3 ms	11.7 Hz	42.7 ms	21.3 ms
8192	170.7 ms	5.9 Hz	85.3 ms	42.7 ms

Table 1: Time and frequency resolution at a sampling rate of 48 kHz.

While the mouse is over the display, a **crosshair (18)** shows the frequency, the note name with its deviation in cents (relative to the reference pitch (21)) and the level in dB at the mouse position, e.g. 441 Hz | A4 +4 ct | -51.8 dB, to follow a melody, to check the intonation or to read the level of a harmonic. A note covers ± 50 cents around its frequency, the same bands as in the keyboard overlay.

A second line shows the **peak**: the strongest maximum of the spectrum in the column under the mouse, within the note band of the mouse (± 50 cents, at least ± 1 bin), e.g. **peak 327.2 Hz | E4 -13 ct | 1.7 dB**. A small white dot marks it, so you always see which maximum the line belongs to; the first line stays the value under the mouse. The frequency and level of the peak are interpolated between the bins (parabola through the three bins around the maximum); with the Hann window the error is a few hundredths of a bin, i.e. below 1 cent above about 200 Hz at FFT size 4096 or 8192. If there is no real maximum in the range (e.g. only the flank of a neighbouring partial, or noise below the colour range), there is no second line and no dot.

Hold **Shift** for the **harmonic cursor**: dashed lines across the display at 2, 3, 4, ... times the frequency of the peak (or of the mouse position if there is no peak), numbered next to the crosshair. You see at once which partials belong to a note, how far the harmonics of a tone reach, or whether a line is a separate tone; deviations show inharmonicity (e.g. piano strings). Lines closer than 4 pixels are left out. Neither the harmonic cursor nor the crosshair is part of the PNG export.

On the linear axis, every frequency bin is drawn exactly at its frequency; where the display shows more bins than it has pixel rows, each row shows the maximum of its bins, so narrow lines do not disappear.

The level values are a power spectral density in dB relative to digital full scale. They are not calibrated with regard to sound pressure level (SPL), and the absolute values depend on the window and the FFT size. Use them to compare levels within the same setting.

The plug-in only analyses: the audio passes through unchanged and without latency.

3.5 What is stored with your project

Your DAW project stores all settings: FFT size, window, overlap, averaging, colourmap, linear/logarithmic axis, display mode, keyboard overlay, BPM grid and its resolution, frequency, colour and time range and the size of the plug-in window (drag the lower right corner to scale it).

4 Further topics

If you want to dive into signal analysis, the spectrogram is just the beginning. I really recommend the web-site of Prof. Julius O. Smith III <https://ccrma.stanford.edu/~jos/sasp/>.

4.1 Source code

This plug-in is open source. You can find the source code here: (<https://github.com/JoergBitzer/JadeSpectrogram2>)

Everything needed to build it is in this repository (clone it with `-recursive` to get JUCE and the TGMStaticLib as submodules, then build it with CMake). The binaries for Windows, macOS and Linux are built automatically by GitHub Actions.

This plug-in belongs to a longer list of plug-ins developed for educational purposes. If you are familiar with JUCE, you will have no problem to understand what's going on. Please report issues at the GitHub repository.

4.2 More Web links

<https://www.musicdsp.org/>

5 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is J. Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT), on macOS the Audio Unit SDK (Apache 2.0), the FFT by Uwe Simmer (MIT-style licence) and colourmap data from matplotlib (CC0); the binaries as a whole are distributed under the AGPLv3. The licence texts are included in the download.

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- Mac[®] is a trademark of Apple Inc.
- VST[®] is a registered trademark of Steinberg Media Technologies GmbH

A Release notes

2.4.6 (October 2026)

- Scrolling display: the image could be shifted against the time axis and the BPM grid when the window was not redrawn after every update (e.g. a covered window). Fixed.

2.4.5 (October 2026)

- The plug-in window could open with the about box showing. Fixed.

2.4.4 (October 2026)

- Source code tidied up (no change in function).

2.4.3 (October 2026)

- BPM grid (synced): the bar line at the loop start was sometimes missing (the host jumps back inside an audio block; found with Cubase). Lines also appear where play starts and after a locate.
- BPM grid (synced): no lines while the transport is stopped, also with block sizes that do not divide the analysis hop (before, a wrong grid could appear).

2.4.2 (October 2026)

- Source code tidied up (no change in function).

2.4.1 (October 2026)

- Faster drawing: the BPM grid needs about 90 % less drawing time (with resolution 1/16 and everything on, a frame takes 30 % less time at the default size); the colour bar is cached. The display looks the same.

2.4.0 (October 2026)

- Peak line in the crosshair readout: the interpolated maximum in the note band of the mouse (frequency, note with cents, level), marked by a white dot.
- Harmonic cursor: hold Shift for lines at the harmonics of the peak (or of the mouse frequency).

2.3.0 (October 2026)

- Reference pitch A4 (380–480 Hz, default 440 Hz) below the frequency axis: note names of the keyboard overlay and of the crosshair follow it.
- The crosshair shows the deviation from the note in cents.
- The version number moved next to the time axis.

2.2.0 (October 2026)

- BPM grid with free tempo: switch between synced (host tempo) and free (own tempo, drag the BPM value), Alt+click in the display sets a bar line. Works also with a stopped transport, while paused and in the Standalone application.
- In synced mode the BPM value follows the host tempo also while the transport is stopped or the display is paused; while paused, a changed host tempo redraws the grid.

2.1.0 (October 2026)

- New overview button (lower right corner): frequency, time and colour range back to their full range.
- The export button moved to the bottom row; the grid resolution box is wider, so $1/16$ is readable.

2.0.1 (October 2026)

- The BPM grid resolution now means note values per bar, as in a DAW grid: 1 = bar lines only, $1/4$ = every quarter note (every beat in $4/4$), $1/16$ = every sixteenth note. Before, it divided the beat, so $1/16$ gave 64 lines per bar in $4/4$. Default is now $1/4$.

2.0.0 (September 2026): the musical spectrogram

New:

- Keyboard overlay: semitone bands and note names over the spectrogram.
- BPM grid: lines at bars, beats and note values (up to $1/16$ note) from the host tempo, exact also after tempo changes, jumps and loops.
- Logarithmic frequency axis (switchable).
- Crosshair with frequency, note and level at the mouse position (replaces the readout below the display).
- Time axis in seconds, 10 s memory (was 8 s) and a time zoom.
- Range sliders for the frequency, colour and time range (drag between the thumbs to move the whole range).
- Averaging along time, overlap 50 % / 75 %.
- Export of the visible spectrogram with its axes as PNG.
- All settings are stored with the project.
- Downloads for Windows, macOS (Universal: Apple Silicon and Intel) and Linux (VST3 and Standalone, on macOS also AU).

Improved:

- Zero latency: the audio passes through unchanged (before, it was delayed by half the FFT size without telling the host).
- Changing FFT size, window or overlap while the audio runs causes no dropouts.
- Linear axis: every frequency bin is drawn exactly at its frequency (before, up to half a bin too high, which shifted the note of the readout at FFT size 2048); where the display has fewer pixel rows than bins, each row shows the maximum, so narrow lines do not disappear.
- Frequency axis labels at round values (e.g. 50, 100 ..., 1.5k).
- Larger window (840×580); the Run/Fix and Lin/Log buttons show what a click will do.
- Based on JUCE 9 (9.0.3); new about box.

Fixed:

- The readout could read past the end of the spectrum at the top of the display.
- The channel mix mode *TimeMean* did not update the spectrum.
- The default colourmap: the selector and the display did not agree (now Plasma).

The changes of version 2.0 were developed with the help of an AI (Anthropic's Claude, in Claude Code) and reviewed and tested by J. Bitzer.

1.1 – 1.2.1 (February 2026): JadeSpectrogram2

- New, self-contained code base: everything needed to build is in one repository, no dependency on Eigen anymore; the analysis uses the fast internal FFT.
- The data exchange between audio and display is block free (FIFO).
- The readout also shows the level in dB.
- More exact scaling.
- Bug fixes: the frequency sliders cannot overlap anymore; negative mouse positions in the readout.

1.0 (2021): JadeSpectrogram

- First release (VST3 and AU), based on the TGM development framework.